

Super-Eddington Little Blue Dots Reionize Helium Too Early

JWST's faint AGN may over-ionize helium unless only a small fraction accrete at extreme rates.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

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Background

JWST found many faint broad-line AGN at $z > 4$, dubbed Little Blue Dots. Some models link them to super-Eddington accretion with strong extreme-UV but weak X-rays.

How does this super-Eddington scenario affect the timing of helium reionization?

What this paper is selling

The authors show that if most LBDs are highly super-Eddington, helium reionization ends too early. Observations require the super-Eddington fraction to be below about 10%.

Key results to remember

- Helium reionization ending after $z \approx 3$ disfavors majority super-Eddington LBDs.
- For fiducial assumptions, super-Eddington fraction must be $\lesssim 10\%$.
- Tighter limits if bright quasars also contribute significantly.

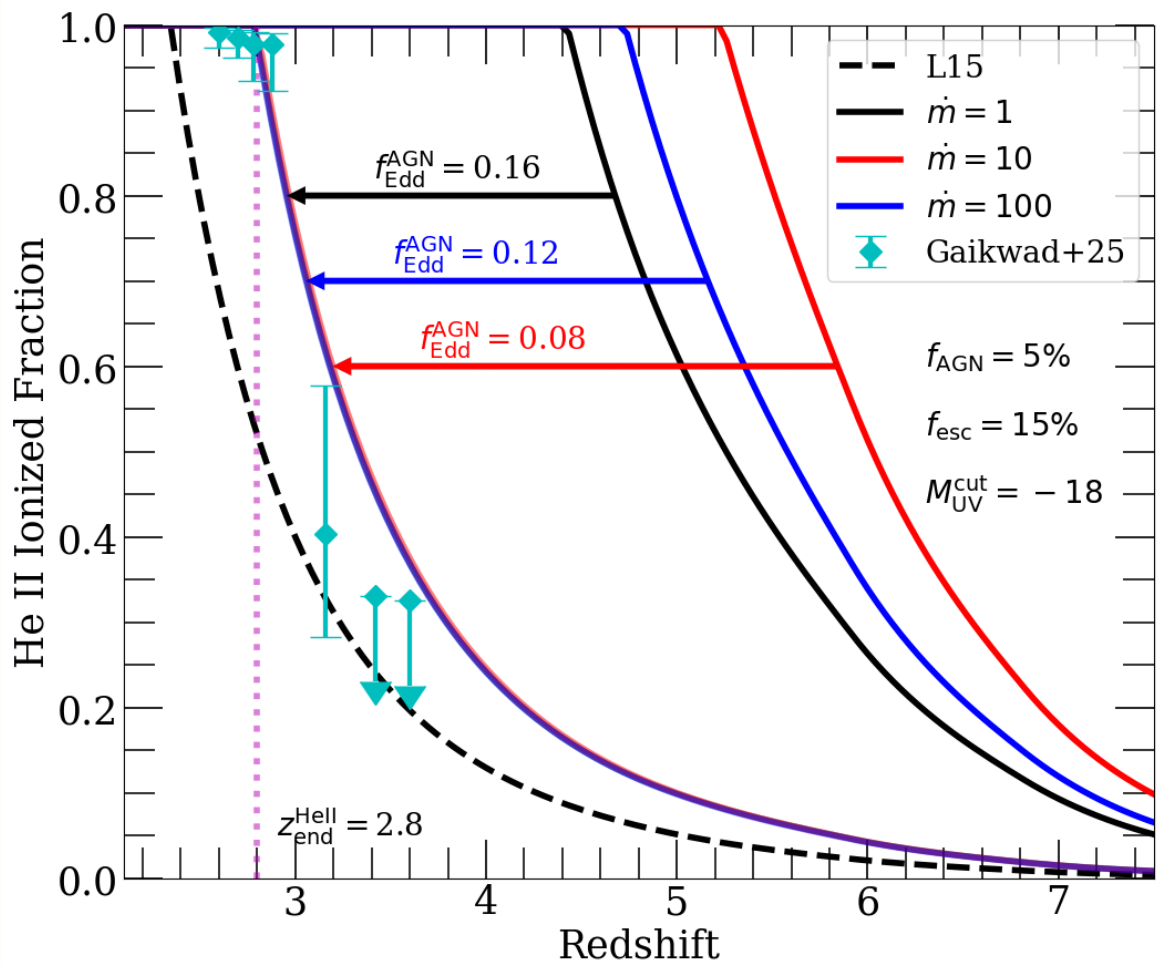


Figure 3 | Helium reionization histories for different SEDs; completing reionization at $z=2.8$ requires specific Eddington fractions.

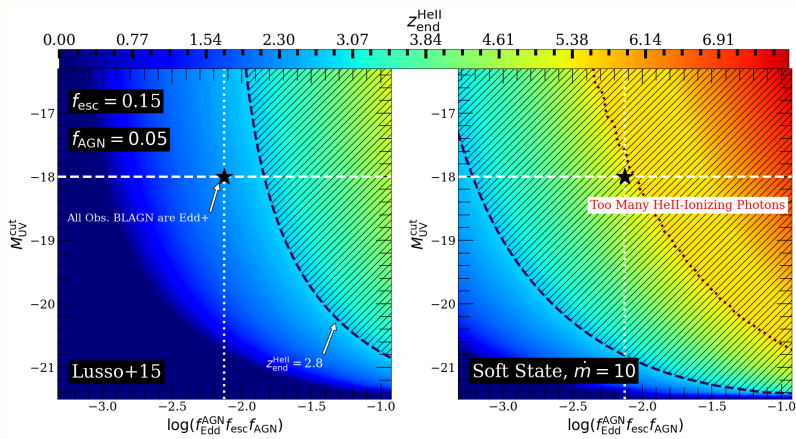


Figure 4 | Excluded AGN parameter space from helium reionization constraints; hatched region shows disfavored high super-Eddington fractions.